Body Copilot

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Figure 1. Fitbit Sense internal circuit board. Bosch BHI160 motion sensor identified in the teardown. (Orange) Source: iFixit, *Fitbit Sense Teardown*, Step 5 [2].

Body CoPilot is using LifeSnaps dataset, which includes Fitbit Sense data from 71 participants across two study rounds. Participants wore the watch on their wrist and were encouraged to continue with their usual daily routines. Data collection lasted for up to 9 weeks in each round. At the end of the study, each participant exported and shared their Fitbit data with the researchers [1]. For this project, we are examining heart rate, heart rate variability (HRV), sleep, steps, and recorded exercise. The Fitbit Sense uses photoplethysmography (PPG) to estimate heart rate and has a three-axis accelerometer for measuring movement. Exercise records include information such as activity type, start time, and duration. We are analyzing measurements and activity records provided by Fitbit rather than collecting measurements ourselves.

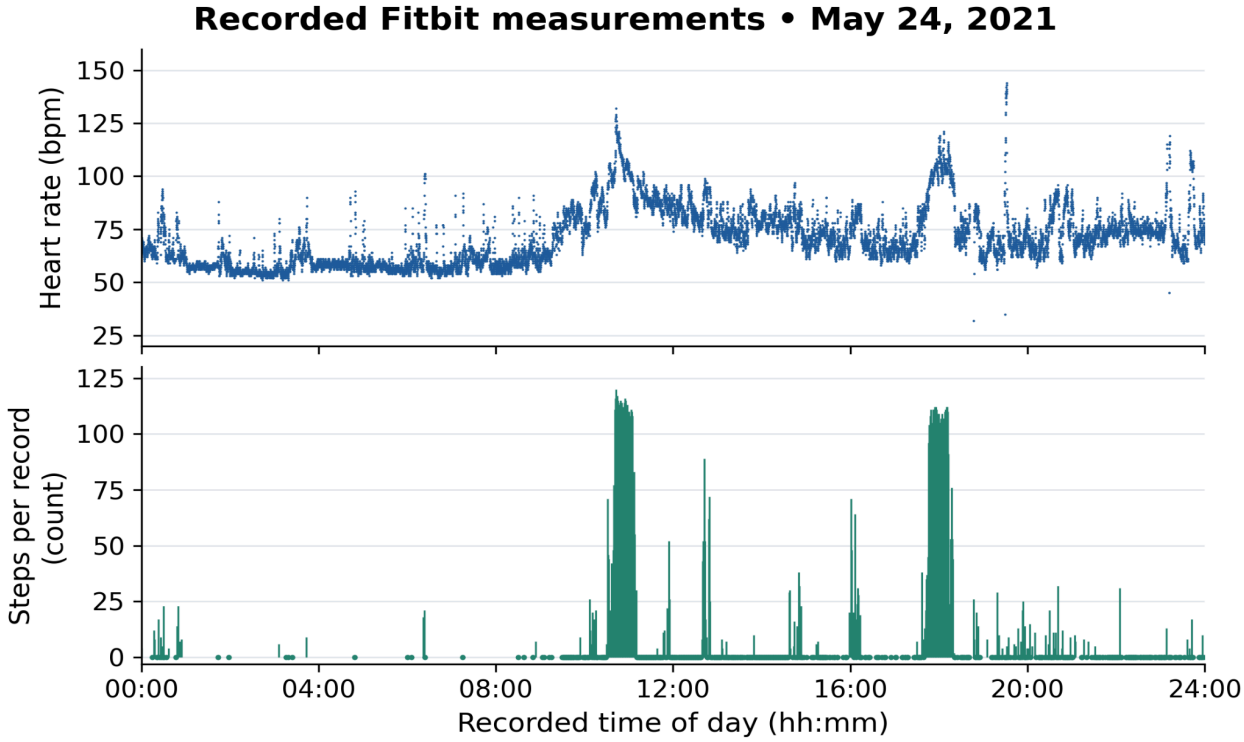


Figure 2. Recorded heart rate (bpm) and steps per record (count) for participant 621e2e8e67b776a24055b564 on May 24, 2021. These are Fitbit outputs.

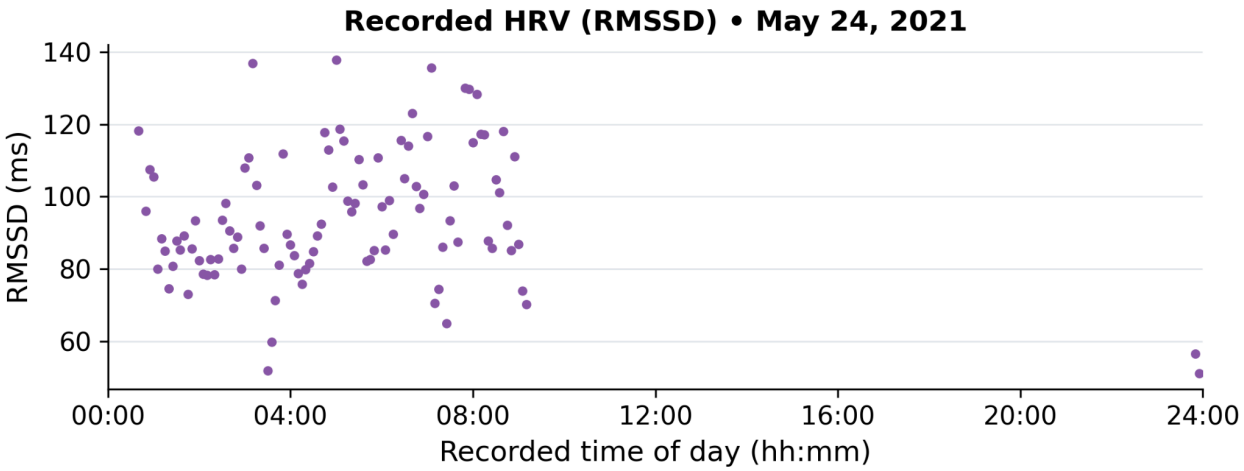


Figure 3. Recorded HRV for the same participant and day.

Measurement	**Most common recording Interval (s)**	**Equivalent Recording Rate (Hz)**
Heart Rate	5	0.2 Hz
Steps	60	0.0167 Hz
HRV	300	0.0033 Hz

These rates were calculated using frequency = 1 / interval in seconds. Longer gaps also occurred, so the records were not evenly spaced. Heart rate estimates were most commonly recorded every 5 seconds showing the rise and fall of heart rate over a certain period of time. Steps records were most commonly spaced 60 seconds apart showing when movement occurs. HRV describes variation between successive heartbeats in milliseconds (ms). This calculation describes variation in heartbeat timing. The 300 second interval in the table describes how often the HRV estimate appears in the sample, not internal sensor sampling frequency. They describe how often Fitbit measurements appear in the data set, rather than how frequently the sensors inside the watch collect original signals. These records help us understand activity within each hour, although missing records can leave gaps.

The Fitbit sense manual content does not include sensor specifications [3]. An iFixit teardown identified a Bosch BHI160 motion sensor inside the Fitbit Sense [2]. Bosch documented accelerometer settings as $\pm$ 2, $\pm$ 4, $\pm$ 8, or $\pm$ 16g with sensitivities of 16,384, 8,192, 4096 and 2,048 counts per g [4]. Range is the minimum and maximum acceleration the sensor can measure. Sensitivity is how many digital counts the output changes per g. Larger ranges capture stronger acceleration but have larger increments, while smaller ranges capture weaker acceleration but have smaller increments.

Bandwidth was documented as 20.25 Hz at a 50 Hz output rate in normal filter mode for the internal BMI160 [5]. The range and filter settings used in LifeSnaps were not confirmed. The optical heart rate sensor's sensitivity, bandwidth, and measurement range have not been verified.

Operating Conditions Accelerometer						
Parameter	**Symbol**	**Condition**	**Min**	**Typ**	**Max**	**Units**
Acceleration Range	g_{FS2g}	Selectable via serial digital interface		±2		g
	g_{FS4g}			±4		g
	g_{FS8g}			±8		g
	g_{FS16g}			±16		g
Start-up time	$t_{A,su}$	Suspend/low power mode to normal mode		3.2		ms

Output Signal Accelerometer						
Parameter	**Symbol**	**Condition**	**Min**	**Typ**	**Max**	**Units**
Resolution				16		bit
Sensitivity	S_{2g}	g_{FS2g}, T_A=25°C		16384		LSB/g
	S_{4g}	g_{FS4g}, T_A=25°C		8192		LSB/g
	S_{8g}	g_{FS8g}, T_A=25°C		4096		LSB/g
	S_{16g}	g_{FS16g}, T_A=25°C		2048		LSB/g

Figure 4. Bosch BHI160/BHI160B datasheet, Table 3 [4].

I examined 68 heart rate readings over a 10 minute period of deep sleep on May 24th, 1:02-1:12 am. Fitbit classified this portion as deep sleep. I calculated heart-rate values, mean heart rate, standard deviation, and minimum and maximum. The mean was 56.79 bpm, the range was 54–59 bpm, and the sample standard deviation was 1.04 bpm. The variation in heart rate can come from normal changes or measurement error and we cannot determine how much came from each. This does not isolate sensor noise. Noise can come from movement including watch movement, wrist flexion or extension, and watch to skin contact.

The Fitbit shines green LED light into the skin, and a photodiode measures the amount of light returning from the tissue. Each heartbeat produces a change in local blood volume, which affects the amount of light returning to the photodiode. Fitbit uses the pattern to estimate beats per minute [6]. The accelerometer measures acceleration along the X, Y, and Z axes. Wrist, arm, and body movement can change these readings. Fitbit's software looks for movement patterns associated with walking and converts them into estimated steps.

Some internal factors are breathing, sleep stages, and exercise, which can cause a change in heart rate or HRV. Some external factors include wrist flexion and extension, arm movement, and a loose watch, which can interfere with readings. Real physiological changes can provide useful information, but measurement errors can make activity patterns harder to identify and reduce Body Copilot's prediction accuracy.

References

[1] LifeSnaps dataset article. Scientific Data (2022).
[2] iFixit. Fitbit Sense Teardown, Step 5.
[3] Fitbit. Fitbit Sense User Manual, Sensors and Components.
[4] Bosch Sensortec. BHI160/BHI160B datasheet, Table 3.
[5] Bosch Sensortec. BMI160 datasheet, Table 12.
[6] Google Fitbit Help. How do I track heart rate with my Fitbit device?